



Green Building Recovery and Reconstruction Guideline

The "Green Building Recovery and Reconstruction Guideline" aims to serve as a brief reference for identifying reconstructed houses that incorporate eco-friendly materials or sustainable construction technologies, promoting environmentally conscious practices.

The criteria outlined below, aligned with GRID (Green, Resilient, and Inclusive Development), are established to assess such houses.

To qualify for a GRID rating, a building must satisfy **any three of the following five** criteria which includes:

1. Walls' Construction Material
2. Roof's Construction Material
3. Passive Design/ Climate Responsive Design Strategies (Energy Efficient measures/solutions)
4. Inclusive
5. Holistic House Design (*Traditional Architecture*)

1. Walls' Construction Material:

According to this criterion, walls are constructed using locally available or environmentally friendly materials as listed in Table 1 such as:

- a. Stone
- b. CSEB Interlocking Bricks
- c. Hollow Fired Bricks/Hollow Concrete Blocks
- d. Bamboo/Timber
- e. Stone Crete
- f. Non-Fired Bricks
- g. Reuse of debris materials
- h. Rammed Earth

2. Roof's Construction Material:

According to this criterion, roofs are constructed based on the locally available and green materials and technologies:

- a. Roof frame/truss construction
 - Treated Timber
 - Treated Bamboo
- b. Materials for roofing
 - Improved Thatch (Refer to NBC 203, 2015)
 - Slate
 - Timber



3. Passive Design/ Climate Responsive Design Strategies (Energy Efficient measures/solutions):

Building should be designed as per the local climatic conditions to ensure a thermal and visual comfortable indoor environment for occupants inside the building. Various climate responsive strategies can be applied to make the building energy efficient as well as comfortable. The measures/solutions should be applied as per the bio-climatic zones which is mentioned below.

a. Walls (External):

Since the wall is the main medium for the entry of heat inside the building or the loss of accumulated heat inside the building, the wall should be constructed in such a way that it has the lowest thermal transmission. For this, the thermal transmission capacity can be controlled by placing different layers on the outer and inner parts of the wall using different energy efficient building materials and technologies.

❖ Energy Efficient Materials and Technologies

- Rat Trap Bond
- Cavity Wall
- Insulated walls, constructed with any walling technique.
- Hollow/Perforated Clay-fired Bricks

(These walls would be called energy efficient, if these technologies are applied to at least 80% of the external wall with above mentioned material or combination of materials.)

Examples of materials with low thermal conductivity: Hollow/Perforated Clay-fired Bricks, AAC Blocks.

b. Roof:

The roofs of the building should have an insulation layer in addition to its timber frame and roofing materials. The construction of attic space should be encouraged in all bio-climatic zones; however, the insulation is necessary in intermediate floor slab separating attic space with lower floors in cool temperate and cold bioclimatic zones to prevent heat loss. The adoption of green roofs can also be encouraged.

Examples of local insulating materials: treated rice husk, hemp, plyboard or timber.

c. Wall openings (Doors and windows)

Doors and windows allow the passage of light and air and are important for a household's energy efficiency because they're how homes let heat and light in and out. Proper window placement and type are essential for energy efficient building design making it a crucial part of the building envelope.

- The ratio of the window to wall ratio (WWR) should not exceed more than 40% in any of the faces in the building.
- The number of north-facing windows should be maximum in warm zones, as the heat accumulation from the north-facing windows is minimal and helps only to extend the daylight and vice versa in cold regions, but with proper overhangs.



- As many openings as possible should be kept towards the north and south façade and openings smaller in size and fewer in numbers should be kept in east and west facades.
- Windows with at least 600mm of overhang should be adopted and mechanisms of external movable shadings in east and west direction should be ensured in warm and temperate regions.
- The additional wooden shutter (from outside) along with glass panes should be adopted in cool temperate and cold regions to avoid heat loss.

d. Natural Ventilation:

Natural ventilation holds significant potential in cooling the indoor environment as well as to maintain indoor air quality. To ensure natural ventilation in the building, minimum openable area must be 1/6th of floor area for warm and temperate climate zone (Ktm Valley By-laws 2064) and 1/16th of the floor area for cool and cold climate zone (NBC:206,2015)

4. INCLUSIVITY:

Within this guideline, households having no other houses anywhere in Nepal or abroad and containing the following members will be deemed eligible.

- Female-Headed HHs,
- Elderly-Headed HHs,
- Child-Headed HHs,
- People with Disability Headed HHs and dependent members in the family.
- Chronically ill-headed HHs
- Low Income Households i.e. monthly income less than 20,000 per a HH with 3 members
- Minorities, Indigenous populations as defined by the government.

5. HOLISTIC HOUSE DESIGN:

- Rainwater Harvesting,
- Solar Panel Installation,
- Improved cooking stove or biogas,
- Grey Water Recycle for agriculture,
- Ventilated Improved Pit Latrine,
- Solid Waste Management: composting,
- Water Efficiency: *One house-one tap*,
- Accessible design for people with disabilities or special needs



Annex:

❖ Table 1: CO₂ emissions for different typology

Construction typology	CO ₂ emissions	Source
Stone masonry in mud mortar (manually and machine dressed stones)	0 kg CO ₂ / m ² (for 18" wall)	<i>Calculated using the calculator developed by BUILDING Energy Efficiency in Nepal (BEEN) Project, 2024</i> <i>The calculation is based on the methodology and protocol prescribed by IPCC.</i>
Hollow concrete block (1:6 cement sand ratio) (for 8" wall)	15.32 kg CO ₂ /m ² 102.13 kg CO ₂ /m ³	
CSEB Interlocking bricks 300x150x100 (for 6" wall)	20.57 kg CO ₂ /m ² 137.16 kg CO ₂ /m ³	
Fired brick Rat Trap Bond (RTB) (1:6 cement sand ratio) (for 9" wall)	31.55 kg CO ₂ /m ² 137.16 kg CO ₂ /m ³	
Hollow/perforated clay-fired bricks, 230 x 110 x 60 mm (3 holes @ 45 mm diameter (1:6 cement sand ratio) (for 9" wall)	38.35 kg CO ₂ /m ² 166.72 kg CO ₂ /m ³	
Fired brick in cement mortar: 240x115x57 mm; (1:6 cement sand ratio for 9" wall	46.37 kg CO ₂ /m ² 201.61 kg CO ₂ /m ³	

**Emissions of transportation are not included in the above calculations.*



- ❖ Table 2: CO₂ emissions recorded by various literatures [It also demonstrate the similar trend of comparisons]

Construction typology	CO ₂ emissions	Source
Hollow concrete block	13 kg CO ₂ / m ² (for 8" wall)	VSBK Project Nepal, 200....
CSEB Interlocking bricks 300x150x100	21 kg CO ₂ / m ² (for 6" wall)	Nepal environment Institute - Comparison of Air Pollutant Emissions between Fired Bricks and Compressed Stabilized Earth Bricks https://bit.ly/3LFvw4u
Fired brick Rat Trap Bond (RTB)	31.47 kg CO ₂ / m ² (for 9" wall)	
Fired brick in cement mortar: 240x115x57	43.87 kg CO ₂ / m ² (for 9" wall)	

**Emissions of transportation are not included in the above calculations.*

- ❖ **Considerations for Passive Design/ Climate Responsive Design Strategies (Energy Efficient measures/solutions):**

Building Orientation: Orient buildings to maximize solar gain during the winter months and minimize exposure during the summer months. In the northern hemisphere, this generally means facing the building southward to capture the most sunlight.

Building Shape and Design: Design buildings with compact shapes to minimize surface area and heat loss. Incorporate features such as large south-facing windows to allow sunlight to enter and heat the interior spaces.

Thermal Mass: Utilize materials with high thermal mass, such as stone or concrete, inside the building. These materials can absorb and store heat during the day and release it gradually at night, helping to stabilize indoor temperatures.

Insulation: Proper insulation is crucial to prevent heat loss during colder months and to keep the interior spaces cool during hot periods. Insulate roofs, walls, and floors effectively to reduce thermal conductivity.



Overhangs and Shading Devices: Design overhangs, awnings, or trellises to provide shading for windows during the summer months when the sun is higher in the sky. This prevents overheating while still allowing sunlight to enter during the winter.

Natural Ventilation: Incorporate features such as operable windows, skylights, and vents to facilitate natural airflow through the building. This helps in cooling the building during warmer months without the need for mechanical ventilation systems.

Greenery and Landscaping: Plant trees, shrubs, and vines strategically to provide additional shading and cooling effects, especially on the western and eastern sides of the building where the sun's angle is lower.

Roof Design: Consider using materials with high reflectivity for roofing to reduce heat absorption. Alternatively, green roofs or roof gardens can provide insulation and cooling benefits while also mitigating stormwater runoff.

Local Materials and Techniques: Utilize locally available materials and traditional building techniques that are suitable for the climate and terrain of the region. This ensures durability, affordability, and compatibility with local architectural styles.

❖ Table 3: Bio- Climate zones of Nepal

S. No.	Climate Zone	Elevation (masl)
1	Warm Temperate (WT)	< 500m
2	Temperate (T)	501m – 1500m
3	Cool Temperate (CT)	1501m – 2500m
4	Cold Climate (C)	> 2500m